# Evaluation runs: zab (zab-broadcast-remix)

generated 2026-11-03 from brain-data b2ae60f86 2026-11-03 17:57:09 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 9a0629e, merge join 367cdc6, JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". $G_{spec}$, $G_{impl}$ = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is $G^{expl}_{spec}$); $G_x$ = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

## stateless

### stateless: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G^{expl}_{spec}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 161 | conforms | 0.0 | 2 | 1 h, 250 GB | 2 | 3 | brain03, 2026-11-02 |
| 2 | 6 | 33161 | conforms | 0.9 | 4 | 1 h, 250 GB | 6 | 21 | brain03, 2026-11-02 |
| 3 | 20 | 9550181 | conforms | 149.9 | 52 | 1 h, 250 GB | 20 | 210 | brain03, 2026-11-02 |
| 4 | | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |

### stateless: pair **issue 1**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G^{expl}_{spec}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 128 | 10 | violation | 0.0 | 2 | 1 h, 250 GB | 12 | 23 | brain03, 2026-11-02 |
| 2 | 6 | 24215 | 70 | violation | 0.6 | 2 | 1 h, 250 GB | 76 | 441 | brain03, 2026-11-02 |
| 3 | 20 | 6881334 | 258 | violation | 105.8 | 36 | 1 h, 250 GB | 278 | 5370 | brain03, 2026-11-02 |
| 4 | | | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |

**Not run (stateless).** planned but never scheduled (grid points beyond the ladder): 3 fifo correct, 3 fifo issue1-propose-to-ackld, 4 correct fifo, 4 fifo correct, 4 fifo issue1-propose-to-ackld, 4 issue1-propose-to-ackld fifo, 5 fifo correct, 5 fifo issue1-propose-to-ackld.

skipped: a smaller size had reached its limit: 5 correct fifo, 5 issue1-propose-to-ackld fifo, 6 correct fifo, 6 issue1-propose-to-ackld fifo, 7 correct fifo, 7 issue1-propose-to-ackld fifo, 8 correct fifo, 8 issue1-propose-to-ackld fifo.

## stateful

### stateful: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 1 | 2 | 161 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2 | 6 | 33161 | conforms | 1.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3 | 20 | 9550181 | conforms | 191.0 | 48 | 48 h, 250 GB | brain05, 2026-11-02 |
| 4 | | | 🕓 (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |

### stateful: pair **issue 1**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 128 | 10 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2 | 6 | 24215 | 70 | violation | 0.7 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3 | 20 | 6881334 | 258 | violation | 121.7 | 41 | 48 h, 250 GB | brain05, 2026-11-02 |
| 4 | | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |

**Not run (stateful).** left out by the table rules (ok): 3 correct fifo, 3 issue1-propose-to-ackld fifo.

skipped: a smaller size had reached its limit: 5 correct fifo, 5 issue1-propose-to-ackld fifo, 6 correct fifo, 6 issue1-propose-to-ackld fifo, 7 correct fifo, 7 issue1-propose-to-ackld fifo, 8 correct fifo, 8 issue1-propose-to-ackld fifo.

planned but never scheduled (grid points beyond the ladder): 4 fifo correct, 4 fifo issue1-propose-to-ackld, 5 fifo correct, 5 fifo issue1-propose-to-ackld, 6 fifo correct, 6 fifo issue1-propose-to-ackld, 7 fifo correct, 7 fifo issue1-propose-to-ackld.

# offline

## offline: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 161 | conforms | 0.3 | 4 | 10 min, 16 GB | 0.2 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 2 | 6 | 33161 | conforms | 1.4 | 4 | 1 h, 250 GB | 0.1 | 1.3 | 0.0 | merge | brain05, 2026-11-03 |
| 3 | 20 | 9550181 | conforms | 192.9 | 50 | 1 h, 250 GB | 0.1 | 192.8 | 0.0 | merge | brain05, 2026-11-03 |

## offline: pair **issue 1**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 128 | 2 | violation | 0.3 | 4 | 10 min, 16 GB | 0.2 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 2 | 6 | 24215 | 5 | violation | 0.9 | 4 | 1 h, 250 GB | 0.1 | 0.8 | 0.0 | merge | brain05, 2026-11-03 |
| 3 | 20 | 6881334 | 9 | violation | 122.0 | 40 | 1 h, 250 GB | 0.1 | 121.8 | 0.0 | merge | brain05, 2026-11-03 |

**Not run (offline).** skipped: a smaller size had reached its limit: 4 correct fifo, 4 issue1-propose-to-ackld fifo, 5 correct fifo, 5 issue1-propose-to-ackld fifo, 6 correct fifo, 6 issue1-propose-to-ackld fifo, 7 correct fifo, 7 issue1-propose-to-ackld fifo, 8 correct fifo, 8 issue1-propose-to-ackld fifo.

planned but never scheduled (grid points beyond the ladder): 3 fifo correct, 3 fifo issue1-propose-to-ackld, 4 fifo correct, 4 fifo issue1-propose-to-ackld, 5 fifo correct, 5 fifo issue1-propose-to-ackld.

# cross offline, Rust implementation against the Java specification

## cross offline, Rust implementation against the Java specification: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 161 | conforms | 3.5 | 160 | 10 min, 250 GB | 3.4 | 0.1 | 0.1 | merge | brain01, 2026-11-03 |
| 2 | 6 | 33161 | conforms | 5.0 | 154 | 10 min, 250 GB | 3.6 | 1.3 | 0.1 | merge | brain01, 2026-11-03 |

## cross offline, Rust implementation against the Java specification: pair **issue 1**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 128 | 2 | violation | 3.5 | 160 | 10 min, 250 GB | 3.4 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 2 | 6 | 24215 | 5 | violation | 4.5 | 154 | 10 min, 250 GB | 3.6 | 0.8 | 0.0 | merge | brain01, 2026-11-03 |

**Not run (cross offline, Rust implementation against the Java specification).** planned but never scheduled (grid points beyond the ladder): 3 fifo correct, 3 fifo issue1-propose-to-ackld, 4 fifo correct, 4 fifo issue1-propose-to-ackld, 5 fifo correct, 5 fifo issue1-propose-to-ackld.

## cross online, Rust implementation (TraceForge) against the Java specification export

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 161 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 2 | 6 | 33161 | conforms | 1.1 | 4 | 1 h, 250 GB | | brain01, 2026-11-02 |

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **issue 1**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 128 | 10 | violation | 0.0 | 4 | 1 h, 250 GB | 2 | brain01, 2026-11-02 |
| 2 | 6 | 24215 | 70 | violation | 0.7 | 4 | 1 h, 250 GB | 2 | brain01, 2026-11-02 |

**Not run (cross online, Rust implementation (TraceForge) against the Java specification export).** planned but never scheduled (grid points beyond the ladder): 3 fifo correct, 3 fifo issue1-propose-to-ackld, 4 fifo correct, 4 fifo issue1-propose-to-ackld, 5 fifo correct, 5 fifo issue1-propose-to-ackld.

## twins offline, Rust implementation against its Java twin

### twins offline, Rust implementation against its Java twin: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 161 | 161 | conforms | 8.3 | 156 | 10 min, 250 GB | 8.1 | 0.1 | 0.1 | merge | brain01, 2026-11-03 |
| 2 | 33161 | 33161 | conforms | 200.7 | 14645 | 10 min, 250 GB | 199.3 | 1.3 | 0.0 | merge | brain01, 2026-11-03 |

### twins offline, Rust implementation against its Java twin: pair **issue 1** against **issue1-propose-to-ackld**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 128 | 128 | conforms | 4.8 | 168 | 10 min, 250 GB | 4.7 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 2 | 24215 | 24215 | conforms | 134.8 | 5916 | 10 min, 250 GB | 134.0 | 0.8 | 0.0 | merge | brain01, 2026-11-03 |

### twins offline, Rust implementation against its Java twin: pair **issue 1**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 161 | 128 | 2 | violation | 8.3 | 156 | 10 min, 250 GB | 8.1 | 0.1 | 0.2 | merge | brain01, 2026-11-03 |
| 2 | 33161 | 24215 | 5 | violation | 200.2 | 14645 | 10 min, 250 GB | 199.3 | 0.8 | 0.0 | merge | brain01, 2026-11-03 |

**Not run (twins offline, Rust implementation against its Java twin).** planned but never scheduled (grid points beyond the ladder): 3 fifo correct, 3 fifo issue1-propose-to-ackld, 4 fifo correct, 4 fifo issue1-propose-to-ackld.

## twins offline, Java implementation against its Rust twin

### twins offline, Java implementation against its Rust twin: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 161 | 161 | conforms | 8.2 | 156 | 10 min, 250 GB | 0.1 | 8.1 | 0.0 | merge | brain01, 2026-11-03 |
| 2 | 33161 | 33161 | conforms | 200.6 | 14645 | 10 min, 250 GB | 1.3 | 199.3 | 0.0 | merge | brain01, 2026-11-03 |

### twins offline, Java implementation against its Rust twin: pair **issue 1** against **issue1-propose-to-ackld**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 128 | 128 | conforms | 4.8 | 168 | 10 min, 250 GB | 0.1 | 4.7 | 0.0 | merge | brain01, 2026-11-03 |
| 2 | 24215 | 24215 | conforms | 135.3 | 5916 | 10 min, 250 GB | 0.8 | 134.0 | 0.5 | merge | brain01, 2026-11-03 |

### twins offline, Java implementation against its Rust twin: pair **issue 1**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 161 | 128 | 2 | violation | 4.8 | 168 | 10 min, 250 GB | 0.1 | 4.7 | 0.1 | merge | brain01, 2026-11-03 |
| 2 | 33161 | 24215 | 5 | violation | 135.2 | 5916 | 10 min, 250 GB | 1.3 | 134.0 | 0.0 | merge | brain01, 2026-11-03 |

**Not run (twins offline, Java implementation against its Rust twin).** planned but never scheduled (grid points beyond the ladder): 3 fifo correct, 3 fifo issue1-propose-to-ackld, 4 fifo correct, 4 fifo issue1-propose-to-ackld.

## context-free twins offline, Rust implementation against the context-free Java port

### context-free twins offline, Rust implementation against the context-free Java port: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 161 | 161 | conforms | 8.8 | 154 | 10 min, 250 GB | 8.7 | 0.1 | 0.1 | merge | brain01, 2026-11-03 |
| 2 | 33161 | 33161 | conforms | 218.8 | 14865 | 10 min, 250 GB | 217.5 | 1.3 | 0.0 | merge | brain01, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **issue 1** against **issue1-propose-to-ackld**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 128 | 128 | conforms | 7.7 | 152 | 10 min, 250 GB | 4.7 | 0.1 | 2.9 | merge | brain01, 2026-11-03 |
| 2 | 24215 | 24215 | conforms | 141.8 | 5783 | 10 min, 250 GB | 140.8 | 0.8 | 0.2 | merge | brain01, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **issue 1**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 161 | 128 | 2 | violation | 8.9 | 154 | 10 min, 250 GB | 8.7 | 0.1 | 0.1 | merge | brain01, 2026-11-03 |
| 2 | 33161 | 24215 | 5 | violation | 218.3 | 14865 | 10 min, 250 GB | 217.5 | 0.8 | 0.0 | merge | brain01, 2026-11-03 |

**Not run (context-free twins offline, Rust implementation against the context-free Java port).** planned but never scheduled (grid points beyond the ladder): 3 fifo correct, 3 fifo issue1-propose-to-ackld, 4 fifo correct, 4 fifo issue1-propose-to-ackld.

## context-free twins offline, Java port against the Rust implementation

### context-free twins offline, Java port against the Rust implementation: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 161 | 161 | conforms | 8.8 | 154 | 10 min, 250 GB | 0.1 | 8.7 | 0.0 | merge | brain01, 2026-11-03 |
| 2 | 33161 | 33161 | conforms | 218.9 | 14865 | 10 min, 250 GB | 1.3 | 217.5 | 0.1 | merge | brain01, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **issue 1** against **issue1-propose-to-ackld**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 128 | 128 | conforms | 4.8 | 152 | 10 min, 250 GB | 0.1 | 4.7 | 0.0 | merge | brain01, 2026-11-03 |
| 2 | 24215 | 24215 | conforms | 141.6 | 5783 | 10 min, 250 GB | 0.8 | 140.8 | 0.0 | merge | brain01, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **issue 1**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 161 | 128 | 2 | violation | 4.8 | 152 | 10 min, 250 GB | 0.1 | 4.7 | 0.1 | merge | brain01, 2026-11-03 |
| 2 | 33161 | 24215 | 5 | violation | 142.3 | 5783 | 10 min, 250 GB | 1.3 | 140.8 | 0.2 | merge | brain01, 2026-11-03 |

**Not run (context-free twins offline, Java port against the Rust implementation).** planned but never scheduled (grid points beyond the ladder): 3 fifo correct, 3 fifo issue1-propose-to-ackld, 4 fifo correct, 4 fifo issue1-propose-to-ackld.

## context-free twins online, Java port (JMC) against the Rust implementation's export

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 161 | 161 | conforms | 9.1 | 159 | 1 h, 250 GB | | brain01, 2026-11-03 |
| 2 | 33161 | 33161 | conforms | 211.0 | 16339 | 1 h, 250 GB | | brain01, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **issue 1** against **issue1-propose-to-ackld**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 128 | 128 | conforms | 8.6 | 157 | 1 h, 250 GB | | brain01, 2026-11-03 |
| 2 | 24215 | 24215 | conforms | 184.1 | 11590 | 1 h, 250 GB | | brain01, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **issue 1**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 161 | 128 | 10 | violation | 8.3 | 173 | 1 h, 250 GB | 105 | brain01, 2026-11-03 |
| 2 | 33161 | 24215 | 70 | violation | 190.4 | 11039 | 1 h, 250 GB | 227 | brain01, 2026-11-03 |

**Not run (context-free twins online, Java port (JMC) against the Rust implementation's export).** planned but never scheduled (grid points beyond the ladder): 3 fifo correct, 3 fifo issue1-propose-to-ackld, 4 fifo correct, 4 fifo issue1-propose-to-ackld.

## context-free twins online, Rust implementation (TraceForge) against the Java port's export

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 161 | 161 | conforms | 0.1 | 4 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 2 | 33161 | 33161 | conforms | 1.1 | 4 | 1 h, 250 GB | | brain01, 2026-11-02 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **issue 1** against **issue1-propose-to-ackld**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 128 | 128 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 2 | 24215 | 24215 | conforms | 0.7 | 4 | 1 h, 250 GB | | brain01, 2026-11-02 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **issue 1**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 161 | 128 | 10 | violation | 0.0 | 4 | 1 h, 250 GB | 1 | brain01, 2026-11-02 |
| 2 | 33161 | 24215 | 70 | violation | 0.7 | 6 | 1 h, 250 GB | 2 | brain01, 2026-11-02 |

**Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export).** planned but never scheduled (grid points beyond the ladder): 3 fifo correct, 3 fifo issue1-propose-to-ackld, 4 fifo correct, 4 fifo issue1-propose-to-ackld.